RAS 585: Machine Learning and Artificial Intelligence (Fall 2026)

Instructor: Shenghan Guo

Assignment 1

This assignment is worth 200 points. It will be due on Wednesday, 10/7/2026, and submitted through Canvas. Code should be written in Python. Other programming languages may be used with the instructor's approval. A typed report (".docx", ".pdf") should be submitted as a separate document along with the code component (".py", ".ipynb") through Canvas.

Data: A dataset, "Raisin_Dataset.csv", is provided to you. A description of the data can be found in "Raisin_Dataset.txt". Please read the document, and then use the data to complete the following tasks.

Note: You may consider each column a variable. The *input attributes* are: "Area", "MajorAxisLength", "MinorAxisLength", "Eccentricity", "ConvexArea", "Extent", "Perimeter". The samples are independent and identically distributed (iid).

Coding component: Your coding component for this assignment should include two parts: (1) Code that you write in a conventional manner without AI assistance. You can use online Python tutorials or coding forums to confirm syntax. No AI is allowed in this part. (2) Code generated by an AI agent. You can choose Codex, Claude, or Gemini. After obtaining the AI-generated code, you will compare your handwritten code with the AI-generated code and then add comments ("#...") in the Python script/notebook generated by AI. Note that these comments should highlight the logical and functional differences between the two versions of code.

Grading: The point value for each part and question is explicitly listed in this assignment. You must make sure that you follow the data, report, and coding guidelines. Failure to submit the required report or coding component will result in a loss of credit. Specifically, missing AI-generated code will result in a 30% deduction, missing handwritten code will result in a 30% deduction, and a missing report will result in a 40% deduction.

Part 1 (70 pts)

1. What are the *classes* in this dataset? (2 pts)

2. Calculate the class-prior log odds for the dataset (this is a scalar number). Write the *discriminant function* in terms of the log odds (this is a formula). (6 pts)

3. Assume that the input attributes are *multivariate normal*. Further assume that the input attributes in each class follow a different multivariate distribution. Calculate the mean vector and covariance matrix for the input attributes in each class. (Hint: consider your answer to Question 1. You should obtain several sets of mean vectors and covariance matrices.) (8 pts)

4. Given your answer to Question 3, generate 20 samples from each of the multivariate distributions. (Hint: the number of samples generated should be 20 times the number of classes.) (10 pts)

5. Given the assumption that the input attributes are multivariate normal, visualize the joint distribution of "MinorAxisLength" and "Perimeter" for each class. Based on the multivariate-normal assumption, do you think that "MinorAxisLength" and "Perimeter" are each univariate normal? Why or why not? (Hint: use your results from Question 3 and visualize the parametric form of the distribution. Create grids for the 3D plots.$[0, 800] \times [0, 800]$) (10 pts)

6. Given your answer to Question 3, write the functional form of the *likelihood ratio*. You may define notation for the mean and covariance of each class. (6 pts)

7. Given your answers to Questions 3 and 6, write the *discriminant function* for each class. Then, calculate the discriminant functions for each sample point and label each of them with the class name. (Hint: see Eq. (4.20) in the textbook. The "label" here is based on your calculated discriminant. You may store the labels in an Excel or .csv file.) (10 pts)

8. Given your answers to Questions 3 and 6, use a pooled covariance matrix for all classes and write the *discriminant function* for each class. Then, calculate the discriminant functions for each sample point and label each of them with the class name. (Hint: see Eqs. (5.21) and (5.22) in the textbook. The "label" here is based on your calculated discriminant. You may store the labels in an Excel or .csv file.) (10 pts)

9. Use a *confusion matrix* to show the classification results obtained using the discriminant functions in Questions 7 and 8. Calculate the classification accuracy for both and compare the results. Briefly describe your findings. (Hint: you will obtain two confusion matrices: one for the results from Question 7 and one for the results from Question 8.) (8 pts)

Part 2 (60 pts)

Do 5-fold cross-validation for the dataset and perform classification analysis. For each of the five folds, do the following:

1. Assume that the input attributes are *multivariate normal*. Calculate the mean vector and covariance matrix for the input attributes in each class using the training data. (10 pts)

2. Given your answer to Question 1, calculate the *discriminant function* for the testing data. Then label each testing sample with the class name. Finally, create a *confusion matrix* to show the classification results for the testing data. (Hint: You may store the labels in an Excel or .csv file.) (15 pts)

3. Given your answer to Question 1, use a pooled covariance matrix for all classes and calculate the *discriminant function* for the testing data. Then label each testing sample with the class name. Finally, create a *confusion matrix* to show the classification results for the testing data. (Hint: You may store the labels in an Excel or .csv file.) (15 pts)

4. For the *discriminant functions* in Questions 2 and 3, respectively, calculate the average *false positive rate*, *false negative rate*, *true positive rate*, and *true negative rate* for the classification results across the five folds. (Hint: you will get four rates for each classification method.) (15 pts)

5. Briefly describe the performance of each discrimination method and identify the best one for this dataset based on the average performance across the five folds of cross-validation. (5 pts)

Part 3 (70 pts)

For this part, take "Area", "MajorAxisLength", "MinorAxisLength", "Eccentricity", "Extent", "ConvexArea" as independent variables, and "Perimeter" as the dependent variable.

1. Visualize "Perimeter" against each independent variable and describe the trends and patterns in your plots. (Hint: you will get 6 plots, each with "Perimeter" as the vertical axis and an independent variable as the horizontal axis.) (8 pts)

2. Use the first 700 samples in the dataset as the training data and the rest as the testing data. Calculate the correlation matrix for the dependent and independent variables for the training data. Based on the

correlation matrix, identify which independent variables have a major association with the dependent variable. Does this association imply a causal relationship? Why or why not? (Hint: Save the correlations in an Excel or .csv file.) (8 pts)

3. Use Python to fit a linear regression model using the training data. Summarize the model coefficients. Based on the coefficients, which independent variables appear to have a greater impact on the dependent variable? (10 pts)

4. Use the model fitted in Question 3 to make predictions for the testing data. Calculate the *mean squared error* for the testing samples with respect to the predictions. Do you think the model has good predictive performance? (Hint: input the testing samples of independent variables into your fitted model and then evaluate the prediction against the true sample values of the dependent variable.) (8 pts)

5. Based on the results from Question 3, do you think that the independent variables are mutually linearly independent? What are the effects of linear dependence among the independent variables on the fitted linear regression model? (5 pts)

6. Perform principal component analysis (PCA) on the training data matrix of independent variables. Show the variance explained by each principal component. (6 pts)

7. Visualize a *Pareto chart* for the variance explained by each principal component. (10 pts)

8. Take the first four principal components from Question 7 and fit a multivariate regression model. Use the model to make predictions for the testing samples and calculate the mean squared error. (Hint: You need to transform the testing data to principal components as well.) (10 pts)

9. Give a practical scenario in which you would use principal component analysis to reduce the data dimensionality before fitting a regression model; give a practical scenario in which you would not use principal component analysis to reduce the data dimensionality before fitting a regression model. (Hint: you can name any data source or situation that is not necessarily related to the Raisin dataset.) (5 pts)